
TEMPO: Regime-Aware Operator Learning with Locally Adapted POD Bases

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Abstract

Operator learning methods approximate the solution map of a parametric PDE with a single global model, an assumption that fails when the governing parameter drives qualitatively distinct solution regimes.

We propose TEMPO, a framework that discovers latent regimes from unlabelled snapshots via Expectation-Maximization, fits a dedicated basis to each, and routes new inputs at inference time. The discovered partition captures both sources of solution variability: regime membership aligns with the shape of the initial condition across all parameter values, while the physical parameter shapes the dynamics within each regime. TEMPO(POD) outperforms the jointly-trained POD-DeepONet baseline by 13–50% on 1D Burgers’ equation and up to $5\times$ on 2D Darcy flow; per-parameter specialists fail catastrophically out-of-distribution.

1 Introduction

Operator learning approximates the solution map of a parametric PDE from data, enabling instant prediction without re-solving the governing equations at each new parameter value [1]. The dominant paradigm — exemplified by DeepONet [2] and POD-DeepONet [3] — trains a single global model over the entire parameter space. This works well when solutions form a simple manifold, but breaks down when the governing parameter drives *qualitatively distinct* dynamics: viscosity separating laminar flow from near-discontinuous shocks, or source magnitude producing solutions that differ by orders of magnitude in amplitude. When regimes differ fundamentally, a single global basis must split its capacity across all of them, fitting none well.

Several methods make the basis richer while still fitting a single model to all trajectories. POD-PoU [4] trains a partition-of-unity mixture of POD bases with a shared branch, reducing error on sharp-gradient problems but without discovering the underlying regime structure. NeuralPOD [5] constructs nonlinear orthogonal basis functions via greedy residual minimization, representing each mode as a neural network and thus overcoming the linear subspace constraint. Neither, however, provides a way to route a new input to the right representation at test time.

Clustering snapshots and fitting local bases to each cluster is a classical idea in reduced-order modeling [6, 7]. Daniel et al. [8] extend this to deep learning by clustering solution subspaces on the Grassmann manifold and training a dictionary selector at inference. These methods assign each snapshot to a single cluster offline, with no feedback between the clustering and the operator being trained on top of it.

We propose **TEMPO** (*Trajectory EM Mixture of POD Operators*), which closes this gap in a principled probabilistic framework. TEMPO models the trajectory ensemble as a Gaussian mixture and recovers regime structure via Expectation-Maximization (EM): the E-step softly assigns each trajectory to regimes based on reconstruction quality, while the M-step rebuilds a dedicated NeuralPOD

basis per regime — so as bases sharpen, assignments improve, and vice versa. After convergence, a GatingNet and per-regime BranchNets are trained jointly, routing new inputs to the correct regime at inference.

Contributions:

- **TEMPO framework.** A two-phase mixture-of-operators for parametric PDEs: an offline EM that discovers M latent regimes from unlabelled snapshots and fits a dedicated orthogonal (POD) or nonlinear (NeuralPOD) basis to each; an online phase where a GatingNet and M BranchNets are trained with a KL penalty that anchors routing to the offline regime partition.
- **Adaptive basis update.** A distribution-shift criterion that monitors moment changes in regime composition across EM iterations and selects per-regime between full retraining, incremental mode addition/pruning, or skipping.
- **Theoretical guarantee.** Regime-local bases achieve reconstruction error no greater than the global rank- K optimum for any soft-assignment matrix and any M, K (Proposition 1).
- **Experiments.** On 1D Burgers' and 2D Darcy flow, TEMPO(POD) reduces error by 13–50% over the jointly-trained POD-DeepONet baseline, while per-parameter specialists fail badly out-of-distribution.

2 Problem Statement

Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain and $\mathcal{T} \subset \mathbb{R}_{\geq 0}$ a time interval.

We are given a dataset of N solution trajectories:

$$\mathcal{D} = \{ (u_0^{(i)}, \kappa^{(i)}, s^{(i)}) \}_{i=1}^N, \quad s^{(i)} = s(\cdot, \cdot; u_0^{(i)}, \kappa^{(i)}) \in L^2(\Omega \times \mathcal{T}), \quad (1)$$

where $u_0^{(i)} \in \mathcal{U} = L^2(\Omega)$ is the initial condition and $\kappa^{(i)} \in \mathcal{K}$ the physical parameter. Each trajectory $s^{(i)}$ is discretized on $N_y = N_t N_x$ quadrature points $\{y_j = (x_j, t_j), w_j\}_{j=1}^{N_y}$ covering the product grid $\Omega \times \mathcal{T}$, with weighted norm $\|f\|_w^2 = \sum_j w_j f(y_j)^2$.

Definition 1 (Solution operator). *The solution operator*

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \rightarrow L^2(\Omega \times \mathcal{T})$$

maps each pair (u_0, κ) to the full space-time solution: $\mathcal{G}(u_0, \kappa) = s(\cdot, \cdot; u_0, \kappa)$.

The operator $\mathcal{G}$ is unknown and expensive to evaluate directly. The *operator learning problem* is to construct $\hat{\mathcal{G}} \approx \mathcal{G}$ from $\mathcal{D}$ alone, generalizing to unseen $(u_0, \kappa) \notin \mathcal{D}$ without solving the PDE.

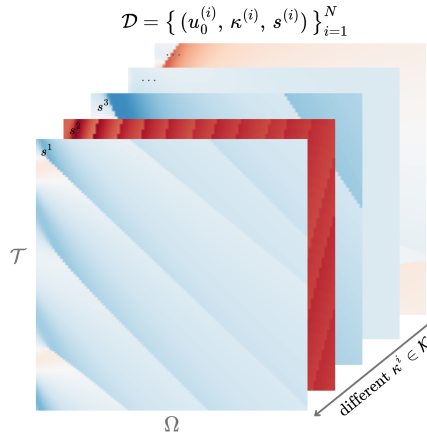


Figure 1: Training dataset $\mathcal{D}$: solution trajectories $s^{(i)}$ on $\Omega \times \mathcal{T}$ for different $\kappa^{(i)} \in \mathcal{K}$. Solutions change character fundamentally across parameter values.

Multi-regime assumption. When solutions change character fundamentally across $\mathcal{K}$, the optimal low-dimensional representation differs from regime to regime, and no single global basis captures the full solution diversity. We formalize this observation via a latent variable model over the trajectory ensemble.

Definition 2 (Generative model). *Each trajectory arises from one of M latent regimes. The regime label $z_i \in \{1, \dots, M\}$ is unobserved and drawn from a categorical prior,*

$$z_i \sim \text{Categorical}(\pi_1, \dots, \pi_M), \quad \pi_m \geq 0, \quad \sum_m \pi_m = 1. \quad (2)$$

Conditioned on $z_i = m$, the trajectory is distributed as

$$(s^{(i)} \mid z_i=m) \sim \mathcal{N}(\hat{s}_m^{(i)}, \sigma^2 W^{-1}), \quad (3)$$

where $W = \text{diag}(w_1, \dots, w_{N_y})$ is the quadrature weight matrix, $\sigma^2 > 0$ controls the softness of regime assignments, and $\hat{s}_m^{(i)}$ is the regime- m approximation to $s^{(i)}$.

The generative model leaves $\hat{s}_m^{(i)}$ unspecified. We require only that it admits a *low-rank trajectory decomposition*:

$$\hat{s}_m^{(i)}(y) = \phi_0^{(m)}(y) + \sum_{k=1}^{K_m} \lambda_k^{(m,i)} \phi_k^{(m)}(y), \quad y = (x, t) \in \Omega \times \mathcal{T}, \quad (4)$$

where $\phi_k^{(m)} : \Omega \times \mathcal{T} \rightarrow \mathbb{R}$ are spatiotemporal mode functions and $\lambda_k^{(m,i)} \in \mathbb{R}$ a scalar amplitude for trajectory i on mode k . This form encompasses POD [3] and learned neural bases [5]; in this work we adopt the latter.

Definition 3 (Regime approximator). *The parameters of regime m are $\Phi_m = \{\phi_k^{(m)}\}_{k=0}^{K_m}$ and scalar amplitudes $\Lambda_m = \{\lambda_k^{(m,i)}\}_{k=1, i=1}^{K_m, N}$, where $\phi_0^{(m)} : \Omega \times \mathcal{T} \rightarrow \mathbb{R}$ is the regime mean field, $\{\phi_k^{(m)}\}_{k \geq 1}$ spatiotemporal mode functions, $\lambda_k^{(m,i)} \in \mathbb{R}$ the scalar amplitude of trajectory i on mode k , and $K_m \leq K_{\max}$ the adaptive mode count.*

The learning problem decomposes into two phases: discovering the latent regime structure from $\mathcal{D}$ *offline*, then training a deployable operator for fast inference *online*.

Phase 1: Offline regime discovery. Since the regime labels z_i in (3) are unobserved, we recover the bases $\{\Phi_m\}$, mixture weights π , and noise level σ^2 by maximizing the observed-data log-likelihood:

$$(\Phi^*, \pi^*, \sigma^{2*}) = \arg \max_{\Phi, \pi, \sigma^2} \sum_{i=1}^N \log \sum_{m=1}^M \frac{\pi_m \det(W)^{1/2}}{(2\pi\sigma^2)^{N_y/2}} \exp\left(-\frac{\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2}{2\sigma^2}\right). \quad (5)$$

The solution yields soft regime responsibilities $\gamma_{im}^* = p(z_i=m \mid s^{(i)}, \Phi^*, \pi^*, \sigma^{2*})$ that serve as supervision for Phase 2.

Phase 2: Online operator learning. With bases $\{\Phi_m^*\}$ and responsibilities $\{\gamma_{im}^*\}$ frozen, we train a gating network (*GatingNet*) and M branch networks (*BranchNets*) that map a new input (u_0, κ) to a weighted combination of local regime predictions. The precise architecture and loss are given in Section 3.4. The quality of the final operator is measured by the mean relative L^2 error:

$$\varepsilon = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{\|s^{(i)} - \hat{s}(\cdot; u_0^{(i)}, \kappa^{(i)})\|_w}{\|s^{(i)}\|_w}. \quad (6)$$

3 Method

All methods studied in this work share a common two-phase structure: an *offline* basis-fitting phase that constructs a low-rank approximation to the solution manifold from snapshots in $\mathcal{D}$, and an *online* operator-learning phase that trains a branch network to predict expansion coefficients from a new input (u_0, κ) . The four methods differ along two independent axes:

- **Basis type.** A *linear* (POD) basis uses SVD of the (weighted) snapshot matrix; a *non-linear* (NeuralPOD) basis learns spatiotemporal mode networks by sequential residual minimization.
- **Number of regimes.** A *single-regime* operator fits one global basis to all snapshots; a *multi-regime* operator (TEMPO) discovers M latent regimes via EM and assigns a dedicated basis to each.

This gives four combinations: POD-DeepONet [3] (linear, global), NeuralPOD-DeepONet (nonlinear, global, our extension of [5]), TEMPO(POD) (linear, M regimes), and TEMPO(NeuralPOD) (nonlinear, M regimes).

3.1 Basis Representations

3.1.1 POD Basis

Let $\phi_0 := \bar{s} = \frac{1}{N} \sum_i s^{(i)}$ be the mean field and $\tilde{S} \in \mathbb{R}^{N_y \times N}$ the matrix with columns $\tilde{s}^{(i)} = s^{(i)} - \phi_0$. The rank- K truncated SVD $\tilde{S} \approx U_K \Sigma_K V_K^\top$ yields orthonormal modes $\{\phi_k\}_{k=1}^K$ (columns of U_K) and coefficients $\lambda_k^{(i)} = [\Sigma_K V_K^\top]_{ki}$; the approximation

$$\hat{s}^{(i)} = \phi_0 + \sum_{k=1}^K \lambda_k^{(i)} \phi_k \quad (7)$$

is optimal in the L^2 sense among all rank- K affine representations.

For TEMPO(POD), the M-step (17) is solved analytically: set $\phi_0^{(m)} := \bar{s}_m = \frac{1}{N_m} \sum_i \gamma_{im} s^{(i)}$, where $N_m = \sum_i \gamma_{im}$, and apply SVD to the weighted centered matrix with columns $\sqrt{\gamma_{im}} (s^{(i)} - \phi_0^{(m)})$. This is the closed-form linear analogue of Algorithm 1.

3.1.2 NeuralPOD Basis

To overcome the linearity constraint, we parametrize each mode function $\phi_k : \Omega \times \mathcal{T} \rightarrow \mathbb{R}$ as a Fourier feature network [9]; we call the resulting basis a *Fourier NeuralPOD* basis:

$$\phi_k(y) = f_{\theta_k}(\psi(y)), \quad \psi(y) = [\sin(2\pi B y), \cos(2\pi B y)], \quad (8)$$

where $B \in \mathbb{R}^{F \times d}$ is a fixed random matrix drawn once at initialization and $f_{\theta_k} : \mathbb{R}^{2F} \rightarrow \mathbb{R}$ is a small MLP with Tanh activations. Plain MLPs exhibit spectral bias [10]; the Fourier encoding resolves this. For multi-scale coverage, the rows of B are partitioned evenly across L frequency scales $\sigma_1 < \dots < \sigma_L$; the ℓ -th block is drawn from $\mathcal{N}(0, \sigma_\ell^2 I)$. The mean field ϕ_0 uses the same architecture.

The scalar amplitudes $\lambda_k^{(i)} \in \mathbb{R}$ are direct learnable parameters, one per training trajectory, optimized jointly with ϕ_k . After Phase 1 they are frozen and serve as regression targets for the BranchNet in Phase 2.

Modes are learned by *greedy residual minimization* [5]. Let $r^{(0,i)} = s^{(i)} - \phi_0$, where ϕ_0 is fitted to the snapshot mean $\bar{s}$ defined above. For each subsequent mode $k \geq 1$, given the residual

$$r^{(k-1,i)}(y) = s^{(i)}(y) - \phi_0(y) - \sum_{\ell=1}^{k-1} \lambda_\ell^{(i)} \phi_\ell(y), \quad (9)$$

we solve

$$\min_{\phi_k, \{\lambda_k^{(i)}\}} \sum_{i=1}^N \sum_j w_j \left(\lambda_k^{(i)} \phi_k(y_j) - r_j^{(k-1,i)} \right)^2 \quad (10)$$

(the global case with uniform weights; the γ -weighted generalization used in TEMPO is Algorithm 1). Mode addition stops when the residual falls below fraction τ of the initial variance,

$$\sum_{i=1}^N \|r^{(k,i)}\|_w^2 < \tau \cdot \sum_{i=1}^N \|s^{(i)} - \bar{s}\|_w^2, \quad (11)$$

or $k = K_{\max}$. Each mode captures the largest remaining variance component — the nonlinear analogue of Gram–Schmidt orthogonalization.

3.2 Single-Regime Operators

3.2.1 POD-DeepONet

POD-DeepONet [3] replaces the learned trunk network of DeepONet [2] with the fixed POD basis. After computing the global modes $\{\phi_k\}_{k=1}^K$ and training coefficients $\{\lambda^{(i)}\}$ offline, a branch network $\mathcal{B}_\theta : \mathbb{R}^{m_s+d_\kappa} \rightarrow \mathbb{R}^K$ is trained online by regression against these coefficients:

$$\mathcal{L}_{\text{branch}} = \frac{1}{N} \sum_{i=1}^N \|\mathcal{B}_\theta(u_0^{(i)}, \kappa^{(i)}) - \lambda^{(i)}\|^2. \quad (12)$$

The prediction at any $(x, t) \in \Omega \times \mathcal{T}$ is

$$\hat{s}(x, t; u_0, \kappa) = \phi_0(x, t) + \sum_{k=1}^K \mathcal{B}_{\theta,k}(u_0, \kappa) \phi_k(x, t). \quad (13)$$

3.2.2 NeuralPOD-DeepONet

Mou et al. [5] introduce NeuralPOD and pair it with a DeepONet branch network to predict mode coefficients for unseen inputs. We adopt this combination, which we refer to as *NeuralPOD-DeepONet*, as a single-regime baseline and as the per-regime basis learner within TEMPO(NeuralPOD).

The procedure follows POD-DeepONet exactly with the Fourier NeuralPOD basis substituted for POD. Offline fitting yields coefficients $\lambda^{(i)} \in \mathbb{R}^K$; online, a branch network minimizes (12) and prediction follows (13). At equal mode count K , the nonlinear basis spans a strictly richer function space than POD.

3.3 TEMPO: Offline Regime Discovery

A single global basis — whether linear or nonlinear — must compromise between qualitatively distinct regimes, as the optimal basis geometry differs fundamentally between them. TEMPO avoids this by constructing a dedicated basis for each of M latent regimes, discovered from unlabelled trajectories via EM on the generative model of Definition 2.

Initialization. A global POD is computed on all N trajectories; each trajectory is projected to a d_0 -dimensional coefficient vector $\alpha^{(i)} \in \mathbb{R}^{d_0}$. Running a standard GMM on $\{\alpha^{(i)}\}$ yields initial responsibilities $\gamma_{im}^{(0)}$, weights $\pi_m^{(0)}$, and per-regime statistics $(\mu_m^{(0)}, \Sigma_m^{(0)})$. Each regime basis $\Phi_m^{(0)}$ is then fitted to the responsibility-weighted ensemble: for TEMPO(NeuralPOD) via NEURALBASIS (Algorithm 1); for TEMPO(POD) via the weighted SVD of Section 3.1. The noise level σ^2 is calibrated from the initial reconstructions rather than treated as a free hyperparameter:

$$\sigma^2 \leftarrow \frac{1}{2N} \sum_{i=1}^N \sum_{m=1}^M \gamma_{im}^{(0)} \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2, \quad (14)$$

σ^2 is held fixed throughout EM; improving bases reduce residuals, which progressively sharpens regime assignments.

E-step. At iteration q , the responsibility of trajectory i for regime m is updated by Bayes' rule on the Gaussian likelihood (3):

$$\gamma_{im}^{(q)} = \frac{\pi_m^{(q-1)} \exp\left(-\frac{\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2}{2\sigma^2}\right)}{\sum_{j=1}^M \pi_j^{(q-1)} \exp\left(-\frac{\|s^{(i)} - \hat{s}_j^{(i)}\|_w^2}{2\sigma^2}\right)}, \quad (15)$$

where $\hat{s}_m^{(i)}$ is evaluated via (4) using $\Phi_m^{(q-1)}$ and $\Lambda_m^{(q-1)}$. As the bases improve in the M-step, the reconstructions $\hat{s}_m^{(i)}$ sharpen and the E-step automatically refines regime assignments in response — distinguishing TEMPO from a GMM applied to a fixed feature space. When κ drives large amplitude

variation (as with $\beta \in [0.01, 100]$ in the Darcy experiments), replacing the squared norm with the relative squared norm $\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2 / \|s^{(i)}\|_w^2$ throughout — in (15), (14), and (17) — keeps E- and M-steps consistent under a heteroscedastic likelihood with noise variance proportional to $\|s^{(i)}\|_w^2$. When κ is observed, the GMM initialization can operate in $\log \kappa$ space, and an optional κ -prior term in the E-step anchors assignments to the known parameter structure.

M-step. With responsibilities fixed, the M-step gives a closed-form update for the mixture weights,

$$\pi_m^{(q)} \leftarrow \frac{N_m^{(q)}}{N}, \quad N_m^{(q)} = \sum_{i=1}^N \gamma_{im}^{(q)}, \quad (16)$$

and a weighted reconstruction problem for each basis:

$$\min_{\Phi_m, \Lambda_m} \sum_{i=1}^N \gamma_{im}^{(q)} \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2, \quad (17)$$

solved by weighted SVD for TEMPO(POD) or NEURALBASIS (Algorithm 1) for TEMPO(NeuralPOD). Since gradient descent solves the NeuralPOD subproblem without guaranteeing global optimality, TEMPO is a Generalized EM algorithm [11], which guarantees that the observed-data log-likelihood is non-decreasing at every iteration.

Proposition 1 (Regime Approximation Advantage). *For any $\{\gamma_{im}\} \geq 0$ with $\sum_m \gamma_{im} = 1$, any $M \geq 1$, any $K \geq 1$:*

$$\varepsilon_{\text{TEMPO}}^K \leq \varepsilon_{\text{global}}^K,$$

where

$$\begin{aligned} \varepsilon_{\text{global}}^K &= \min_{\mu} \sum_i \|(I-P)(s^{(i)} - \mu)\|_w^2, \\ \varepsilon_{\text{TEMPO}}^K &= \sum_m \min_{\mu_m} \sum_i \gamma_{im} \|(I-P_m)(s^{(i)} - \mu_m)\|_w^2 \end{aligned}$$

are the global and regime-wise optimal rank- K reconstruction errors. *Proof: Appendix A.1.*

Remark 1. Both $\varepsilon_{\text{TEMPO}}^K$ and $\varepsilon_{\text{global}}^K$ use exactly K basis functions per sample at inference; TEMPO stores M regime-specific bases of rank K offline, but routes each sample through exactly one.

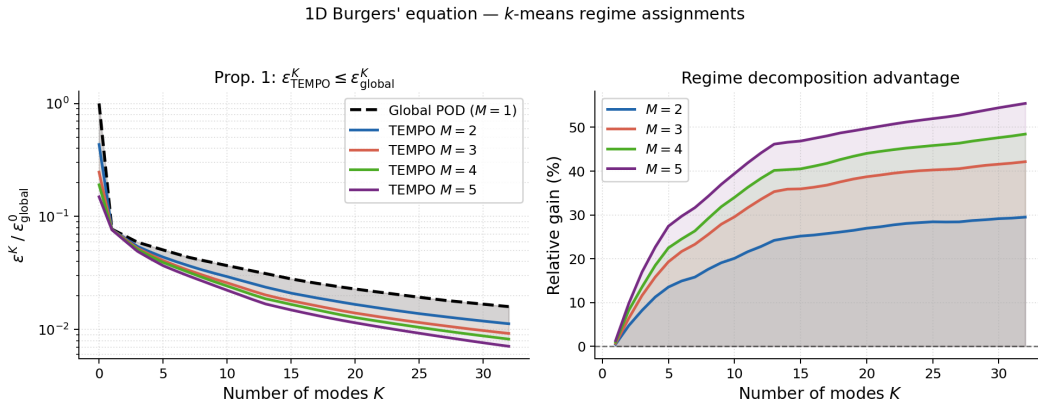


Figure 2: Empirical verification of Proposition 1 on 1D Burgers'. *Left:* normalized reconstruction error (log scale); TEMPO lies strictly below the global POD for all $K \geq 1$. *Right:* relative gain $(\varepsilon_{\text{global}}^K - \varepsilon_{\text{TEMPO}}^K) / \varepsilon_{\text{global}}^K$; all values are non-negative, with gains of 20–39% at $K=10$.

Adaptive basis update. Retraining all M NeuralPOD bases at every EM iteration is computationally prohibitive. We introduce a distribution-shift criterion to determine whether each basis requires

updating. Let $\alpha^{(i)}$ denote the global POD projection of $s^{(i)}$, computed once during initialization and fixed thereafter. The responsibility-weighted first and second moments of regime m are

$$\mu_m^{(q)} = \frac{1}{N_m^{(q)}} \sum_{i=1}^N \gamma_{im}^{(q)} \alpha^{(i)}, \quad (18)$$

$$\Sigma_m^{(q)} = \frac{1}{N_m^{(q)}} \sum_{i=1}^N \gamma_{im}^{(q)} (\alpha^{(i)} - \mu_m^{(q)}) (\alpha^{(i)} - \mu_m^{(q)})^\top. \quad (19)$$

The distribution-shift criterion measures the relative change in these moments between consecutive iterations:

$$\Delta_m^{(q)} = \frac{\|\mu_m^{(q)} - \mu_m^{(q-1)}\|_2}{\|\mu_m^{(q-1)}\|_2 + \epsilon} + \frac{\|\Sigma_m^{(q)} - \Sigma_m^{(q-1)}\|_F}{\|\Sigma_m^{(q-1)}\|_F + \epsilon}. \quad (20)$$

where $\epsilon > 0$ is a small numerical stabilizer. $\Delta_m^{(q)}$ measures how much the composition of regime m has changed, not just the volume of reassignments. Two trajectories with very different $\alpha^{(i)}$ swapping regimes produce a large $\Delta_m^{(q)}$ and trigger a basis update; two similar snapshots doing so do not. Given thresholds $0 < \varepsilon_s < \varepsilon_l$, regime m is SKIPPED if $\Delta_m^{(q)} < \varepsilon_s$, updated INCREMENTALLY (add or prune one mode) if $\varepsilon_s \leq \Delta_m^{(q)} < \varepsilon_l$, and fully retrained from scratch (FULL RERUN) otherwise. In the INCREMENTAL case, we compute the residual of the current basis under the updated responsibilities $\gamma^{(q)}$, add a new mode if $\sum_i \gamma_{im}^{(q)} \|r_m^{(K_m, i)}\|_w^2 \geq \tau \cdot \sum_i \gamma_{im}^{(q)} \|s^{(i)} - \bar{s}_m^{(q)}\|_w^2$, and prune the last mode if its weighted contribution falls below ε_p . The FULL RERUN case uses a cold-start random initialization, as warm-starting may trap gradient descent in the geometry of the previous regime.

Convergence and model selection. The number of regimes M is selected by two criteria: BIC on the EM log-likelihood (penalizes complexity) and mean assignment entropy H . When they conflict, entropy is decisive: a large gain ΔH from M to $M+1$ signals a new regime. EM terminates when $\max_m \Delta_m^{(q)} < \varepsilon_c$. Selected values are $M=3$ (Burgers’), $M=4$ (Navier-Stokes), and $M=5$ (Darcy); details for Burgers’ are in Table 4 (Appendix A.3). The complete offline procedure is given in Algorithm 2.

3.4 TEMPO: Online Gated Operator

After Algorithm 2 converges, all regime bases $\{\Phi_m^*\}$ and responsibilities $\{\gamma_{im}^*\}$ are frozen. The online phase trains a GatingNet $G : \mathbb{R}^{m_s+d_\kappa} \rightarrow \Delta^{M-1}$ that maps (u_0, κ) to regime weights, and M BranchNets $\mathcal{B}_m : \mathbb{R}^{m_s+d_\kappa} \rightarrow \mathbb{R}^{K_m}$ that predict expansion coefficients for each regime. The TEMPO prediction at query point $y = (x, t) \in \Omega \times \mathcal{T}$ is

$$\hat{s}(y; u_0, \kappa) = \sum_{m=1}^M g_m(u_0, \kappa) \left[\phi_0^{(m)}(y) + \sum_{k=1}^{K_m} b_{m,k}(u_0, \kappa) \phi_k^{(m)}(y) \right], \quad (21)$$

where $\mathbf{g} = G(u_0, \kappa)$ and $\mathbf{b}_m = \mathcal{B}_m(u_0, \kappa)$. The BranchNet outputs $\mathbf{b}_m$ predict the expansion coefficients $\lambda_m^{(i)}$ for new inputs (u_0, κ) not seen in training. The prediction is mesh-free and evaluable at any $y \in \Omega \times \mathcal{T}$ without retraining.

Training objective. Writing $g_m^{(i)} = g_m(u_0^{(i)}, \kappa^{(i)})$, the online loss is

$$\mathcal{L}_{\text{online}} = \underbrace{\frac{1}{N} \sum_{i=1}^N \|s^{(i)} - \hat{s}(\cdot; u_0^{(i)}, \kappa^{(i)})\|_w^2}_{\mathcal{L}_{\text{data}}} + \underbrace{\eta_{\text{KL}} \frac{1}{N} \sum_{i,m} g_m^{(i)} \log \frac{g_m^{(i)}}{\gamma_{im}^*}}_{\mathcal{L}_{\text{KL}}} - \underbrace{\eta_H \frac{1}{N} \sum_{i,m} g_m^{(i)} \log g_m^{(i)}}_{-\mathcal{L}_{\text{ent}}}. \quad (22)$$

$\mathcal{L}_{\text{data}}$ drives reconstruction accuracy. $\mathcal{L}_{\text{KL}}$ anchors the online gating to the offline regime partition: it penalizes deviation of $\mathbf{g}^{(i)}$ from the EM responsibilities γ_i^* , so the network cannot discard the discovered structure in pursuit of a simpler routing. $\mathcal{L}_{\text{ent}}$ prevents collapse to a single expert, ensuring all M regime bases contribute. The hyperparameters $\eta_{\text{KL}}, \eta_H \geq 0$ trade off these objectives.

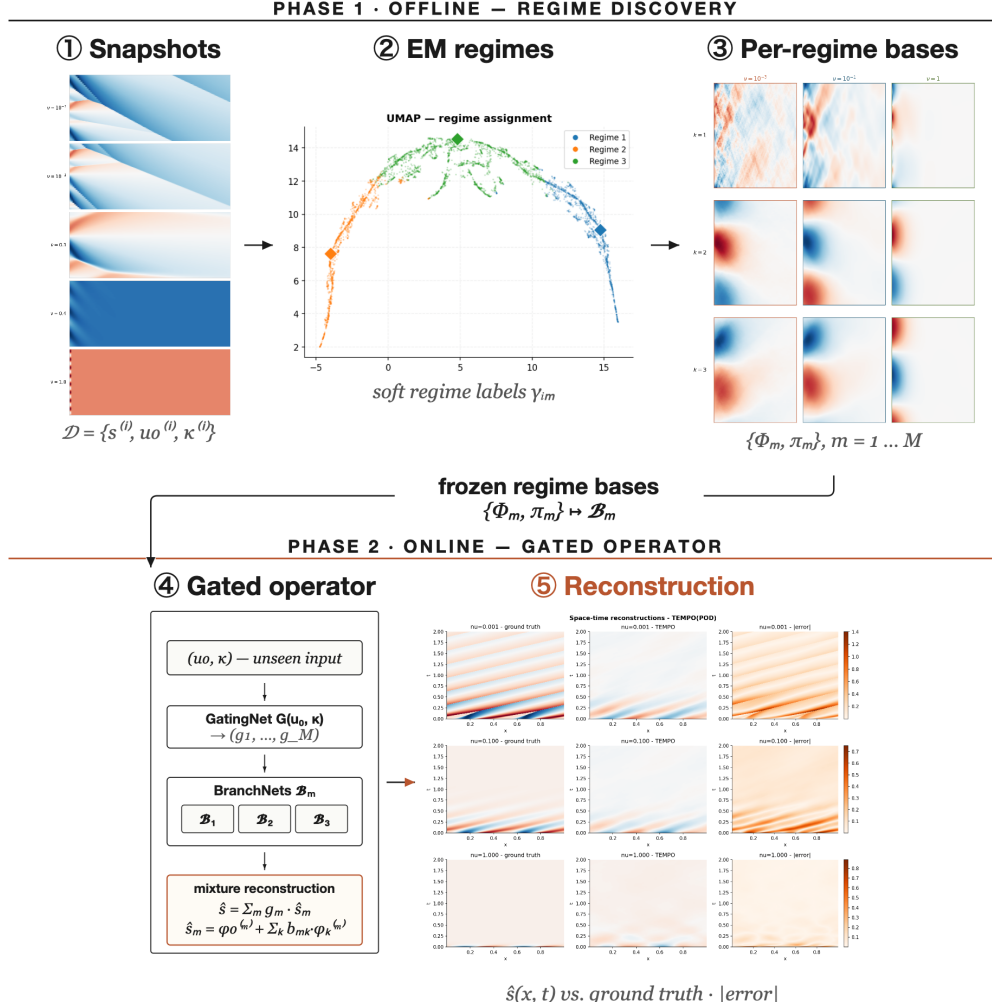


Figure 3: **TEMPO pipeline** (illustrated on 1D Burgers’). Offline: EM discovers M regimes, each with a dedicated basis $\{\Phi_m\}$. Online: a GatingNet and M BranchNets produce a weighted reconstruction.

4 Computational Experiments

We evaluate TEMPO on parametric PDE benchmarks, verifying three claims: (i) the offline EM phase recovers interpretable, physics-consistent regime structure from unlabelled snapshots; (ii) locally adapted bases achieve lower reconstruction error than a single global basis at equal mode count; (iii) a gated operator with per-regime bases outperforms a single global basis when the data contains structured regimes.

4.1 Experimental Setup

Datasets. We use three parametric PDE benchmarks; the first two are from PDEBench [12].

1D Burgers’ equation.

$$u_t + u u_x = \nu u_{xx}, \quad x \in (-1, 1), \quad t \in (0, 2], \quad (23)$$

with periodic boundary conditions and random initial conditions. Viscosity $\nu \in \{0.001, 0.01, 0.1, 1.0\}$ spans four qualitatively distinct solution regimes: smooth diffusive profiles at $\nu=1.0$ and near-discontinuous shock structures at $\nu=0.001$. Each value provides $N=9,500$ trajectories on $N_x=1,024$ spatial points and $N_t=201$ time steps (38,000 total).

2D Darcy flow.

$$-\nabla \cdot (a(x) \nabla u(x)) = \beta, \quad x \in (0, 1)^2, \quad (24)$$

with zero Dirichlet boundary conditions. Here $a(x)$ is a spatially varying permeability field sampled from a random sum of Gaussians, and $\beta \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ is the constant source term. Each value provides $N=8,000$ training and 1,000 test samples discretised on a 128×128 uniform grid ($N_y=16,384$ spatial points), 40,000 samples in total. The operator maps $a(x)$ to the pressure solution $u(x)$; as a steady-state problem, $N_t=1$ and there is no time dependence.

2D incompressible Navier-Stokes.

$$\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \frac{1}{\text{Re}} \Delta \omega + f, \quad \mathbf{x} \in (0, 1)^2, \quad t \in (0, 2], \quad (25)$$

with periodic boundary conditions, $\omega = \nabla \times \mathbf{u}$, divergence-free $\mathbf{u}$ recovered via the stream function, and Kolmogorov-type forcing $f(\mathbf{x}) = 0.1 (\sin 2\pi(x_1+x_2) + \cos 2\pi(x_1+x_2))$ [13]. Reynolds number $\text{Re} \in \{100, 1000, 3600, 10000\}$ spans four qualitatively distinct regimes from smooth laminar flow to developed turbulence. Each value provides $N=5,000$ trajectories on a 64×64 periodic grid (20,000 total; 4,000/1,000 train/test); see Appendix A.2. Operator: $\mathbf{u}_0 \mapsto (\mathbf{u}_t)_{t=1}^{20}, \Delta t=0.1$.

Methods. We compare DeepONet [2], POD-DeepONet [3], FNO [13], and CNO [14]—all trained jointly on all parameter values via a single global model. We introduce NeuralPOD-DeepONet (learned Fourier NeuralPOD basis with branch network); TEMPO(POD) and TEMPO(NeuralPOD) (regime-aware bases discovered via EM, routed by gating network). Per-parameter specialists serve as performance ceilings. All methods implemented by the authors. Evaluation: mean relative L^2 error (6) per parameter value. Inference times measured on a single NVIDIA RTX A5000 GPU, per sample.

4.2 Main Results

1D Burgers’ equation. Table 1 gives the cross-viscosity error matrix. Per- ν specialists are unviable across the full range: a $\nu=0.1$ model degrades from 0.048 to 0.461 at $\nu=0.001$. TEMPO(POD) addresses this by discovering regime structure jointly, outperforming the joint POD-DeepONet baseline by 13–50% across all ν . TEMPO(NeuralPOD) does not surpass the joint baseline. NeuralPOD-DeepONet already underperforms POD-DeepONet in the single-regime case (0.229 vs. 0.048 at $\nu=0.1$): the branch network finds NeuralPOD coefficients harder to predict than POD projections. In TEMPO this is amplified: each EM iteration fits M NeuralPOD bases from scratch, and errors in early iterations feed back through the responsibilities into later ones.

2D Darcy flow. Table 2 gives the cross- β matrix. Specialist failure is catastrophic — Darcy solutions scale linearly with β , driving cross-parameter errors beyond 1000. TEMPO(POD) reduces error by 3–5 $\times$ over the joint POD-DeepONet baseline (e.g., $\beta=0.1$: 0.775 vs. 2.557; $\beta=100$: 0.073 vs. 0.347). TEMPO(NeuralPOD) improves substantially over the global NeuralPOD-DeepONet baseline ($\beta=0.1$: 3.813 vs. 72.9) but does not match TEMPO(POD). FNO and CNO do not decompose solutions into explicit bases and are not directly comparable; TEMPO’s value lies in discovering interpretable regime structure, not in raw accuracy.

2D Navier-Stokes. Table 3 gives the cross-Re error matrix. TEMPO(POD) matches the joint POD-DeepONet baseline within 5–17% across all Re but does not improve over it. On this benchmark, solutions at different Re share enough structure for a single global basis to be competitive, and dedicated per-regime bases bring no clear gain. TEMPO(NeuralPOD) (κ -initialised, $M=4$) converges but remains worse than TEMPO(POD).

4.3 Phase 1 Reconstruction Advantage

Figure 2 empirically verifies Proposition 1 on 1D Burgers’ ($N=1,500$, $\nu \in \{0.001, 0.1, 1.0\}$) with k -means regime assignments. TEMPO achieves 20–39% lower reconstruction error than the global POD at $K=10$ modes, with the gap growing in both M and K ; no violations of the bound were observed across all $K \in \{1, \dots, 32\}$.

Table 1: **Relative L^2 test error on 1D Burgers’ equation.** Each per- ν specialist is trained on a single viscosity and evaluated on all three jointly-trained values; underlined entries are in-distribution. The large off-diagonal errors motivate joint training with regime discovery. All jointly-trained methods use $\nu \in \{0.001, 0.1, 1.0\}$. **Bold:** best per column among jointly-trained methods.

Method	Train ν	$\nu=0.001$	$\nu=0.1$	$\nu=1.0$	Infer. (ms)
<i>Per-ν specialists (single-viscosity training)</i>					
DeepONet [2]	0.001	<u>0.536</u>	0.474	0.382	0.037
	0.1	0.535	<u>0.471</u>	0.349	
	1.0	0.547	0.483	<u>0.283</u>	
POD-DeepONet [3]	0.001	<u>0.273</u>	0.297	0.639	0.004
	0.01	0.274	0.274	0.614	
	0.1	0.349	<u>0.048</u>	0.344	
	1.0	0.461	<u>0.260</u>	<u>0.025</u>	
NeuralPOD-DeepONet (<i>ours</i>)	0.001	<u>0.392</u>	0.288	0.535	3.53
	0.01	0.391	0.284	0.541	
	0.1	0.426	<u>0.229</u>	0.371	
	1.0	0.483	<u>0.315</u>	<u>0.183</u>	
FNO [13]	0.001	<u>0.426</u>	0.529	0.623	9.90
	0.01	0.454	0.401	0.492	
	0.1	0.650	<u>0.304</u>	0.521	
	1.0	1.161	0.919	<u>0.158</u>	
CNO [14] (2.0M)	0.01	0.687	0.955	2.344	2.14
	0.1	0.983	<u>0.199</u>	0.471	
	1.0	1.870	1.084	<u>0.125</u>	
<i>Joint training ($\nu \in \{0.001, 0.1, 1.0\}$)</i>					
DeepONet [2]	all	0.533	0.472	0.289	0.037
POD-DeepONet [3]	all	0.460	0.339	0.210	0.004
FNO [13]	all	0.427	0.298	0.202	9.90
CNO [14] (2.0M)	all	0.361	0.162	0.133	2.14
TEMPO(POD) (<i>ours</i>)	$M=2$	0.319	0.150	0.159	0.025
	$M=3$	0.317	0.168	0.182	
TEMPO(NeuralPOD) (<i>ours</i>)	$M=2$	0.536	0.458	0.326	0.024
	$M=3$	0.542	0.456	0.288	

4.4 Regime Discovery

We select $M=3$ based on entropy gain: H increases by $+0.30$ from $M=2$ to $M=3$ but only $+0.18$ from $M=3$ to $M=4$, indicating the third regime captures genuine structure that a fourth does not. BIC increases monotonically with M (favoring $M=2$ by complexity alone), so entropy is the decisive criterion here (Table 4, Appendix A.3). Figure 4 confirms three distinct manifold branches, one per discovered regime.

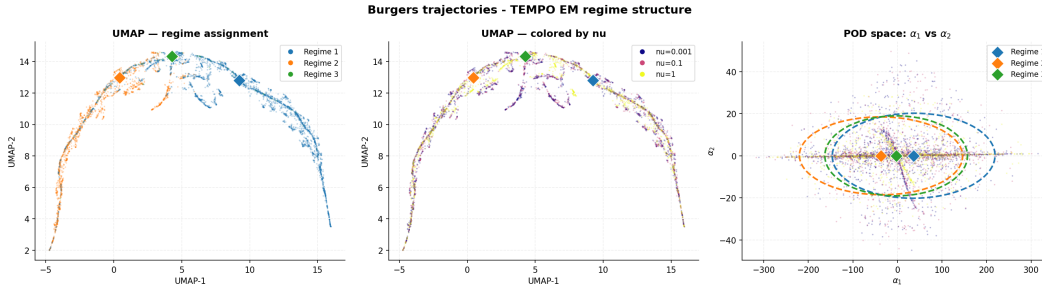


Figure 4: TEMPO(POD) regime structure on 1D Burgers’ ($M=3$, 20 EM iterations). *Left:* UMAP colored by regime. *Center:* same embedding colored by ν . *Right:* GMM fit in POD coefficient space (α_1, α_2); stars mark regime centroids μ_m .

Table 2: **Relative L^2 test error on 2D Darcy flow.** Each per- β specialist is trained on a single source value and evaluated on all five; underlined entries are in-distribution. Off-diagonal errors exceed 1 because the Darcy solution scales with β , so a specialist trained on $\beta=100$ predicts amplitudes $\sim 10^4$ times larger than the ground truth at $\beta=0.01$. TEMPO trains jointly on $\beta \in \{0.1, 1, 10, 100\}$; the $\beta=0.01$ regime is excluded from joint training (—). **Bold:** best per column among jointly-trained methods.

Method	Train β	$\beta=0.01$	$\beta=0.1$	$\beta=1.0$	$\beta=10$	$\beta=100$	Infer. (ms)
<i>Per-β specialists (single-value training)</i>							
DeepONet [2]	0.01	4.197	0.615	0.944	0.994	0.999	0.025
	0.1	>10	<u>0.851</u>	0.875	0.987	0.999	
	1.0	>10	>1	<u>0.433</u>	0.902	0.972	
	10	>100	>10	>1	<u>0.243</u>	0.705	
	100	>1000	>100	>10	>1	<u>0.078</u>	
POD-DeepONet [3]	0.01	<u>0.449</u>	0.791	0.961	0.996	>1	0.001
	0.1	>1	<u>0.148</u>	0.871	0.986	0.999	
	1.0	>10	>1	<u>0.048</u>	0.897	0.990	
	10	>100	>10	>1	<u>0.035</u>	0.900	
	100	>1000	>100	>10	>1	<u>0.042</u>	
NeuralPOD-DeepONet (<i>ours</i>)	0.01	<u>1.407</u>	0.786	0.962	0.996	>1	0.168
	0.1	>1	<u>0.421</u>	0.902	0.989	0.999	
	1.0	>10	>1	<u>0.135</u>	0.901	0.990	
	10	>100	>10	>1	<u>0.050</u>	0.900	
	100	>1000	>100	>10	>1	<u>0.037</u>	
FNO [13]	0.01	<u>0.529</u>	0.791	0.967	0.997	0.9998	0.587
	0.1	6.26	<u>0.144</u>	0.877	0.988	0.999	
	1.0	73.1	8.41	<u>0.034</u>	0.909	0.992	
	10	>100	72.6	8.18	<u>0.013</u>	0.900	
	100	>1000	256.3	43.2	6.13	<u>0.008</u>	
CNO [14] (2.0M)	0.01	<u>0.388</u>	0.929	1.000	1.001	1.000	2.24
	0.1	10.6	<u>0.125</u>	0.948	1.004	1.002	
	1.0	67.0	7.84	<u>0.024</u>	0.896	0.989	
	10	>1000	99.1	8.37	<u>0.009</u>	0.898	
<i>Joint training ($\beta \in \{0.1, 1, 10, 100\}$)</i>							
DeepONet [2]	all	—	76.53	8.503	1.352	0.493	0.025
POD-DeepONet [3]	all	—	2.557	0.363	0.181	0.347	0.001
NeuralPOD-DeepONet (<i>ours</i>)	all	—	72.9	6.92	0.357	0.345	0.168
FNO [13]	all	—	0.380	0.083	0.023	0.008	0.587
CNO [14] (2.0M)	all	—	0.133	0.026	0.008	0.005	2.24
TEMPO(POD) (<i>ours</i>)	$M=5$	—	0.775	0.178	0.136	0.073	0.006
TEMPO(NeuralPOD) (<i>ours</i>)	$M=5$	—	3.813	0.549	0.322	0.311	0.006

Each regime corresponds to a geometric type of initial condition u_0 that appears across all viscosity values. The per-viscosity gating weights (Table 5, Appendix A.3) are nearly uniform across $\nu \in \{0.001, 0.1, 1.0\}$, closely matching the global weights $\pi=(0.324, 0.267, 0.409)$: regime membership correlates with initial-condition geometry across all viscosity values. Within each regime, ν shapes the dynamics: the same initial geometry yields a shock at $\nu=0.001$ and a smooth wave at $\nu=1.0$. At inference, ν is passed to the branch network as a conditioning input.

4.5 Role of the Physical Parameter

We ablate the effect of κ at inference by replacing it with zero or a learned prediction. On Burgers', TEMPO(NeuralPOD) barely needs κ : removing it costs only 1–3% in L2, since u_0 alone carries enough information to route inputs, consistent with regimes being defined by initial-condition shape. TEMPO(POD) is more sensitive (+37%); however, ν can be predicted from u_0 with 87% accuracy, reducing the gap to +7%. On Darcy, κ is essential: removing β collapses gating to a fixed mixture and increases error by 2–3 $\times$; β cannot be predicted from the permeability field $a(x)$ because it

Table 3: **Relative L^2 test error on 2D incompressible Navier-Stokes.** Each per-Re specialist is trained on a single Reynolds number and evaluated on all four; underlined entries are in-distribution. Joint methods train on all Re simultaneously. **Bold:** best per column among jointly-trained methods. [†] kappa-initialised variant.

Method	Train Re	Re=100	Re=1000	Re=3600	Re=10000	Infer. (ms)
<i>Per-Re specialists (single-Re training)</i>						
DeepONet [2]	100	<u>1.000</u>	1.002	1.003	1.004	0.135
	1000	1.000	<u>1.000</u>	1.000	1.000	
	3600	1.002	1.000	<u>1.000</u>	1.000	
	10000	1.001	1.000	1.000	<u>1.000</u>	
POD-DeepONet [3]	100	<u>0.082</u>	0.748	0.865	0.877	0.005
	1000	0.667	<u>0.129</u>	0.671	0.812	
	3600	0.660	0.392	<u>0.180</u>	0.623	
	10000	0.934	0.687	0.556	<u>0.345</u>	
NeuralPOD-DeepONet (<i>ours</i>)	1000	0.762	<u>0.198</u>	0.607	0.761	4.87
FNO [13]	100	<u>0.013</u>	0.262	0.368	0.437	0.30
	1000	0.287	<u>0.033</u>	0.138	0.227	
	3600	0.381	<u>0.150</u>	<u>0.039</u>	0.128	
	10000	0.461	0.260	0.126	<u>0.042</u>	
CNO [14]	100	<u>0.014</u>	0.047	0.055	0.058	0.50
	1000	0.027	<u>0.028</u>	0.033	0.035	
	3600	0.030	<u>0.029</u>	<u>0.033</u>	0.035	
	10000	0.030	0.028	0.032	<u>0.034</u>	
<i>Joint training (all Re)</i>						
POD-DeepONet [3]	all	0.092	0.132	0.141	0.145	0.005
FNO [13]	all	0.009	0.019	0.024	0.027	0.30
NeuralPOD-DeepONet (<i>ours</i>)	all	0.700	0.713	0.314	0.320	4.87
CNO [14]	all	0.020	0.035	0.040	0.043	0.50
TEMPO(POD) (<i>ours</i>)	$M=4$	0.097	0.144	0.155	0.170	0.021
TEMPO(NeuralPOD) [†] (<i>ours</i>)	$M=4$	0.595	0.620	0.694	0.627	0.021

is drawn independently in the dataset (classifier accuracy 21%, below chance). Full results are in Table 6 (Appendix A.3).

5 Conclusion

Operator learning for parametric PDEs typically relies on a single global representation. When the governing parameter drives qualitative change in solution structure, a single basis must fit incompatible geometries and prediction quality suffers.

TEMPO discovers regime structure from unlabelled snapshots via EM and fits a dedicated basis to each regime. The discovered partition captures both sources of solution variability: on 1D Burgers', regimes group solutions by the shape of the initial condition across all viscosity values, while viscosity shapes the dynamics within each regime. The same initial condition produces a shock at $\nu=0.001$ and a smooth wave at $\nu=1.0$, yet belongs to the same regime throughout. A global model conflates these two sources of variation; TEMPO separates them without supervision.

TEMPO(POD) outperforms the joint POD-DeepONet baseline by 13–50% on Burgers' and up to $5\times$ on Darcy flow, where per-parameter specialists fail catastrophically out-of-distribution. On Navier-Stokes, TEMPO matches the POD baseline. TEMPO(NeuralPOD) extends the framework to nonlinear bases; closing the gap with TEMPO(POD) remains an open direction.

Several directions remain open. Conditioning basis functions on κ would let each mode encode parameter-dependent dynamics directly. Extending regime discovery to continuous parameter spaces and three-dimensional problems is a natural next step.

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A Appendix

A.1 Proof of Proposition 1

Lemma 1. Let $\bar{s} = \frac{1}{N} \sum_i s^{(i)}$ be the global mean and $\bar{s}_m = \frac{1}{N_m} \sum_i \gamma_{im} s^{(i)}$ the regime- m mean, with $N_m = \sum_i \gamma_{im}$. For any rank- K orthoprojector P and any regime m ,

$$\sum_i \gamma_{im} \|(I-P)(s^{(i)} - \bar{s}_m)\|_w^2 = \sum_i \gamma_{im} \|(I-P)(s^{(i)} - \bar{s})\|_w^2 - N_m \|(I-P)(\bar{s}_m - \bar{s})\|_w^2.$$

Proof. Let $c_m = \bar{s}_m - \bar{s}$. Writing $(I-P)(s^{(i)} - \bar{s}_m) = (I-P)(s^{(i)} - \bar{s}) - (I-P)c_m$ and expanding $\|a - b\|_w^2 = \|a\|_w^2 - 2\langle a, b \rangle_w + \|b\|_w^2$, then summing over i with weights γ_{im} :

$$\begin{aligned} \sum_i \gamma_{im} \|(I-P)(s^{(i)} - \bar{s}_m)\|_w^2 &= \sum_i \gamma_{im} \|(I-P)(s^{(i)} - \bar{s})\|_w^2 \\ &\quad - 2 \left\langle (I-P) \sum_i \gamma_{im} (s^{(i)} - \bar{s}), (I-P)c_m \right\rangle_w \\ &\quad + N_m \|(I-P)c_m\|_w^2. \end{aligned}$$

Since $\sum_i \gamma_{im} (s^{(i)} - \bar{s}) = N_m c_m$, the cross-term equals $-2N_m \|(I-P)c_m\|_w^2$; together with the quadratic term this gives $-N_m \|(I-P)c_m\|_w^2 = -N_m \|(I-P)(\bar{s}_m - \bar{s})\|_w^2$. $\square$

Proof of Proposition 1. Let P^* and P_m^* be the globally and regime- m optimal rank- K projectors. Since P_m^* minimizes the regime- m objective over all rank- K projectors, substituting the (generally suboptimal) P^* cannot decrease it:

$$\varepsilon_{\text{TEMPO}}^K = \sum_m \sum_i \gamma_{im} \|(I-P_m^*)(s^{(i)} - \bar{s}_m)\|_w^2 \leq \sum_m \sum_i \gamma_{im} \|(I-P^*)(s^{(i)} - \bar{s}_m)\|_w^2.$$

Applying Lemma 1 to each m and summing over m , then using $\sum_m \gamma_{im} = 1$ for each i :

$$\begin{aligned} \sum_m \sum_i \gamma_{im} \|(I-P^*)(s^{(i)} - \bar{s}_m)\|_w^2 &= \sum_i \|(I-P^*)(s^{(i)} - \bar{s})\|_w^2 - \underbrace{\sum_m N_m \|(I-P^*)(\bar{s}_m - \bar{s})\|_w^2}_{\geq 0} \\ &\leq \sum_i \|(I-P^*)(s^{(i)} - \bar{s})\|_w^2 = \varepsilon_{\text{global}}^K. \end{aligned}$$

$\square$

A.2 2D Navier-Stokes Data Generation

We use the same equation and forcing as Li et al. [13] with a pseudo-spectral solver in stream-function formulation: diffusion treated implicitly (Crank–Nicolson), advection explicitly (second-order Adams–Bashforth), and $\frac{2}{3}$ -rule dealiasing on the 64×64 grid. Simulation time-step $\delta t = 0.01$; each recorded snapshot aggregates 10 sub-steps ($\Delta t = 0.1$). Initial vorticity fields are random sums of sinusoids,

$$\omega_0(\mathbf{x}) = \sum_{k=1}^7 \frac{a_k}{1+k} [\sin(2\pi k x_1 + \phi_k) + \cos(2\pi k x_2 + \phi_k)], \quad a_k \sim \mathcal{N}(0, 1), \quad \phi_k \sim \mathcal{U}[0, 2\pi].$$

A 5 s burn-in (500 simulation steps) discards transient behavior before recording begins.

A.3 Additional Experiments

Regime model selection (1D Burgers’). Table 4 reports BIC, mean assignment entropy H , and mean test error $\bar{\varepsilon}$ for $M \in \{2, 3, 4\}$. Table 5 shows the per-viscosity EM responsibilities alongside global mixture weights π_m .

Table 4: Regime count selection on 1D Burgers’. BIC = $-2 \log \mathcal{L} + k \log N$ (lower is better; penalizes complexity); H : mean assignment entropy (higher gain signals a new genuine regime); $\bar{\varepsilon}$: mean relative L^2 test error. BIC increases monotonically, favoring $M=2$ by complexity alone. **Bold row**: selected configuration ($M=3$), chosen because the entropy gain $\Delta H=0.30$ from $M=2 \rightarrow 3$ substantially exceeds $\Delta H=0.18$ from $M=3 \rightarrow 4$.

M	BIC ($\times 10^4$, $\downarrow$)	H	$\bar{\varepsilon}$
2	3.90	0.622	0.209
3	4.92	0.920	0.222
4	5.47	1.099	0.228

Table 5: Mean EM assignment weights per viscosity (1D Burgers’, TEMPO(POD), $M=3$) vs. global mixture weights π_m . Each row sums to 1.

ν	Regime 1	Regime 2	Regime 3
0.001	0.32	0.28	0.40
0.1	0.32	0.26	0.42
1.0	0.33	0.26	0.41
π_m	0.32	0.27	0.41

Role of the physical parameter (ablation). Table 6 reports the effect of replacing κ with zero or a learned prediction at inference.

Table 6: κ ablation: increase in mean relative L2 error when κ is removed or predicted, relative to the real- κ baseline. For Darcy, β is drawn independently of $a(x)$; prediction from input data is not possible.

	Burgers’		Darcy	
	NeuralPOD ($M=3$)	POD ($M=3$)	NeuralPOD ($M=5$)	POD ($M=5$)
$\kappa = 0$	+0.006 (+1%)	+0.127 (+37%)	+0.334 (+27%)	+0.618 (+219%)
κ predicted from u_0	—	+0.024 (+7%)	not predictable	

POD vs. NeuralPOD basis comparison (1D Burgers’). Figure 5 shows the singular-value and residual decay for POD ($K=30$ modes capture 99.99% of the variance) and Fourier NeuralPOD. The corresponding spatiotemporal mode functions (Figure 6) reveal a structural difference: POD modes are globally supported, while NeuralPOD modes localise adaptively — at $\nu=0.01$ they concentrate near the shock front. Per-sample reconstructions from both bases are compared in Figure 8; additional NeuralPOD examples are shown in Figure 7.

Representative reconstructions (1D Burgers’). Figure 9 shows space–time predictions for all four methods on 1D Burgers’ ($\nu=1.0$). Each panel: ground truth (*left*), prediction (*center*), point-wise error (*right*).

TEMPO regime structure across benchmarks. Figure 10 compares the Phase 1 regime structure across all three benchmarks. On Burgers’ (NeuralPOD), the partition mirrors the POD variant (Figure 4) with sharper assignments. On Darcy flow, regimes span multiple β values; TEMPO(NeuralPOD) finds a nearly identical partition. On Navier-Stokes, all Re values overlap in POD coefficient space and regimes 2–4 collapse to a few points, explaining why dedicated bases bring no gain.

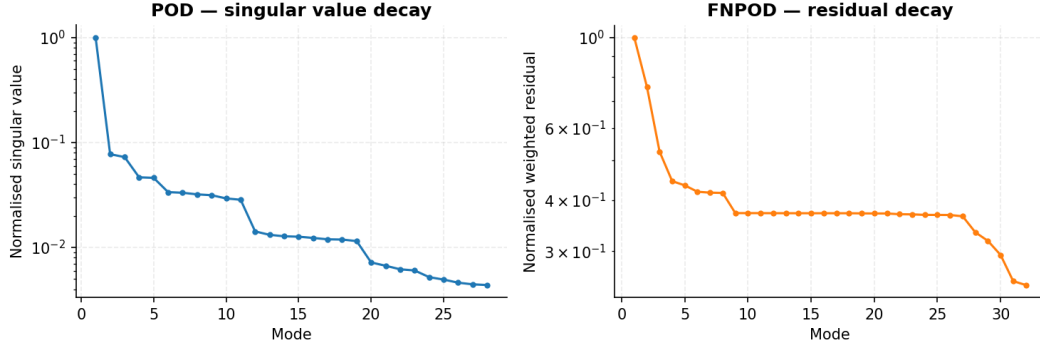


Figure 5: Basis convergence on 1D Burgers' ($\nu=1.0$): POD singular-value decay (*left*); Fourier NeuralPOD residual decay (*right*).

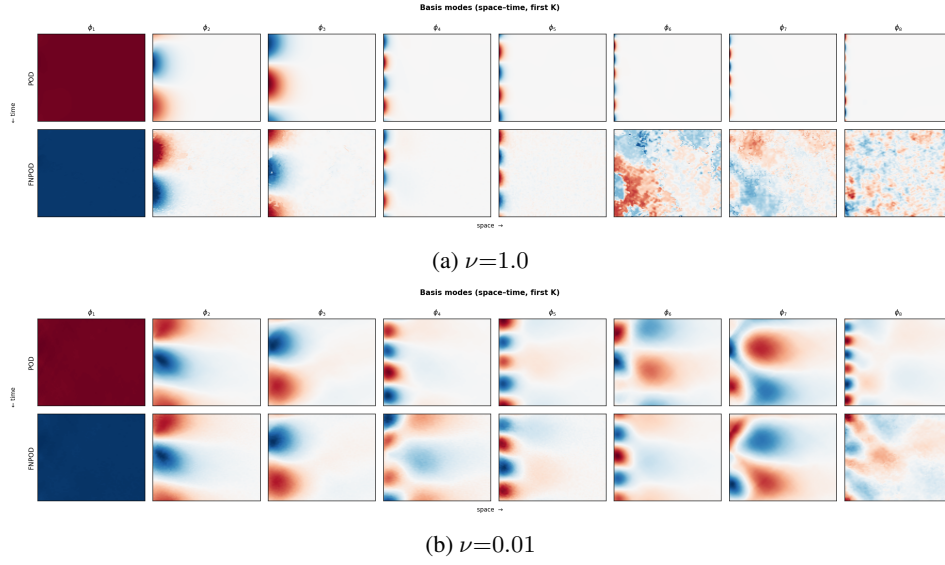


Figure 6: First K basis modes on 1D Burgers': POD (*top row*) vs. Fourier NeuralPOD (*bottom row*).

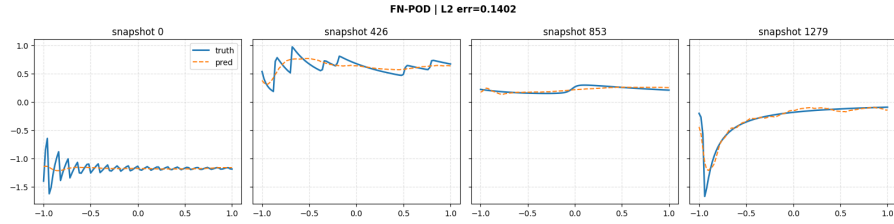


Figure 7: Fourier NeuralPOD representative reconstructions on 1D Burgers' ($\nu=1.0$).

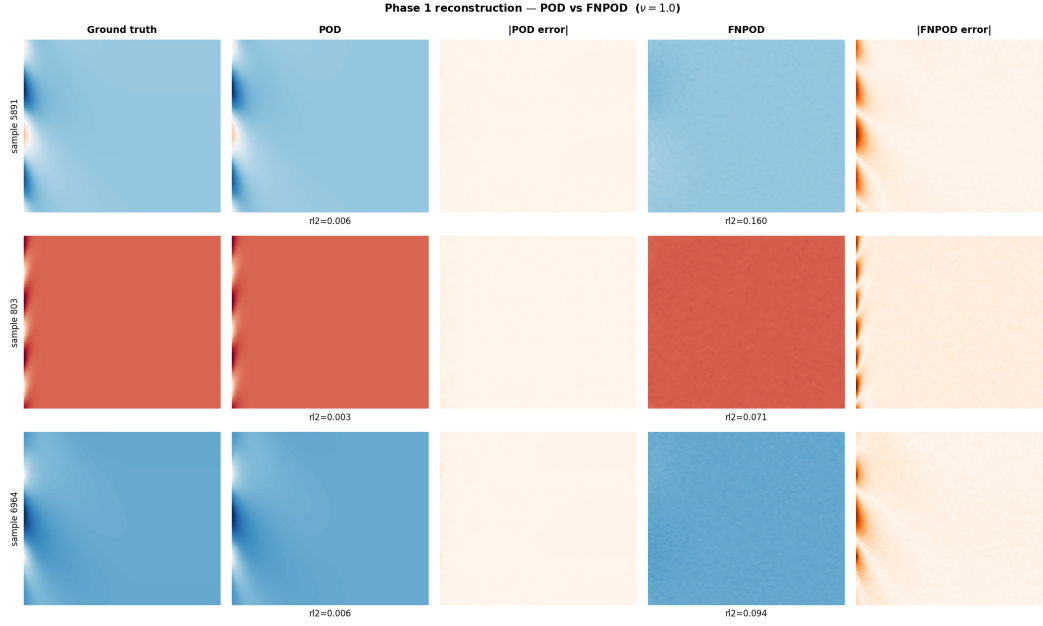


Figure 8: Phase 1 reconstruction comparison on 1D Burgers' ($\nu=1.0$). Each row shows one sample: ground truth, POD reconstruction, |POD error|, Fourier NeuralPOD reconstruction, |FNPOD error|.

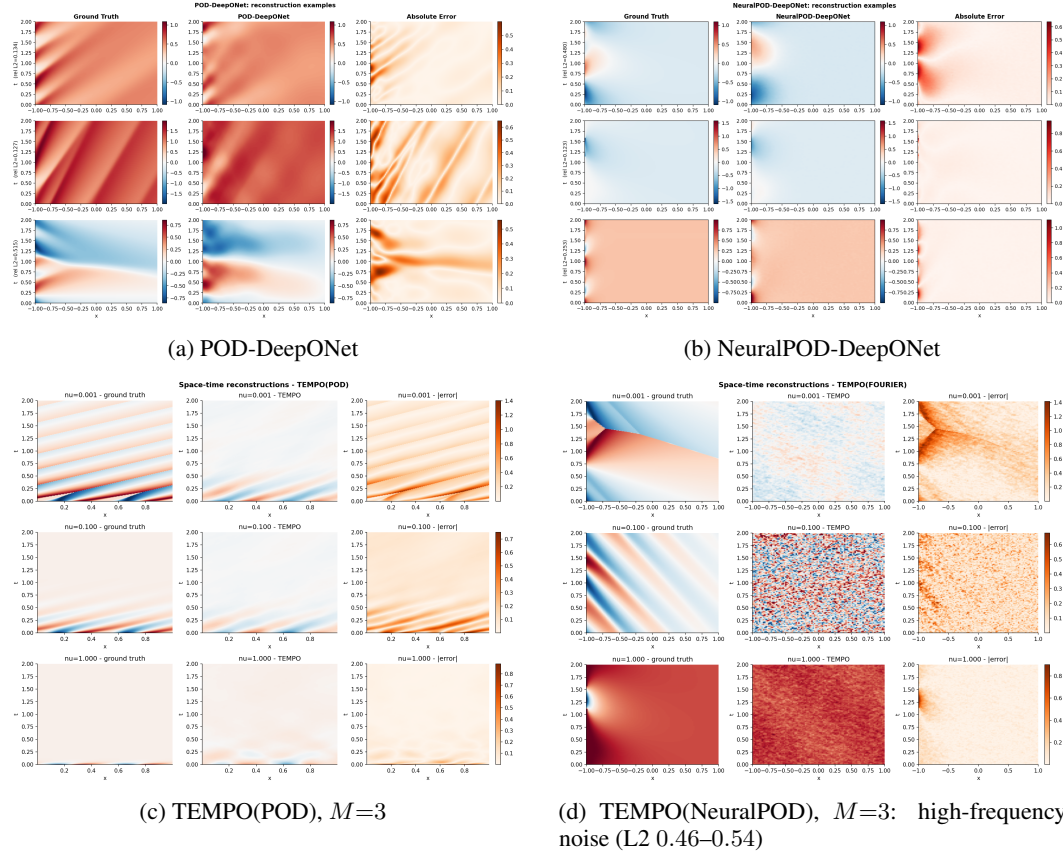
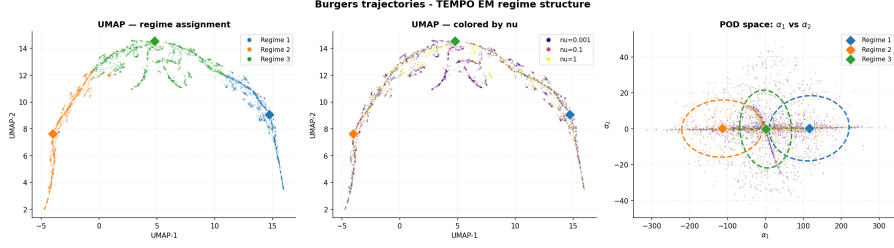
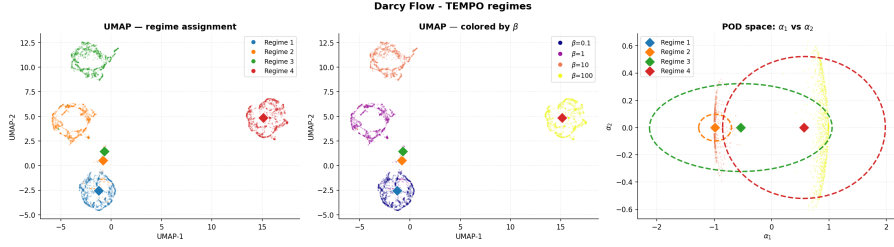


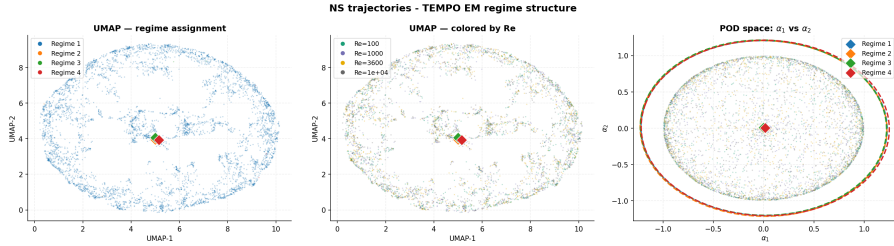
Figure 9: Space-time reconstructions on 1D Burgers' ($\nu=1.0$).



(a) 1D Burgers', NeuralPOD ($M=3$, $H=0.23$)



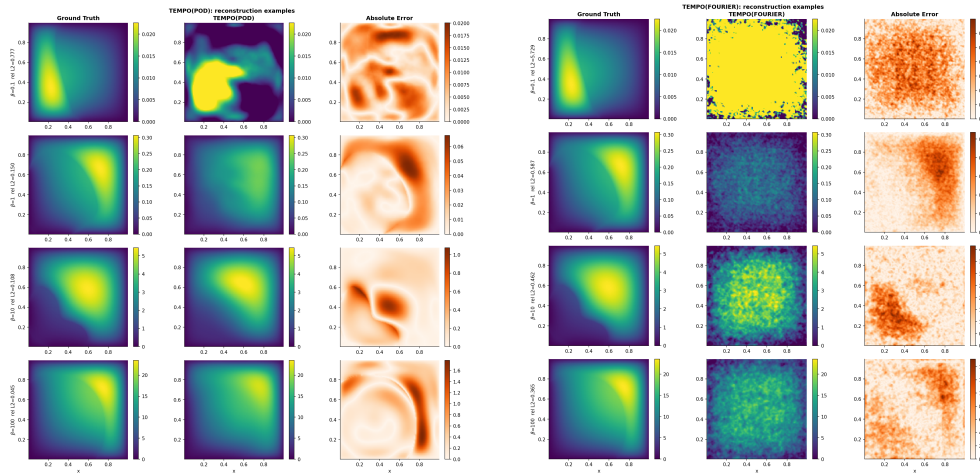
(b) 2D Darcy flow, POD ($M=4$)



(c) 2D Navier-Stokes, POD ($M=4$): no Re separation

Figure 10: TEMPO EM regime structure across all benchmarks. Layout as in Figure 4.

TEMPO reconstructions (2D Darcy flow). Figure 11 shows representative predictions from both TEMPO variants on 2D Darcy flow ($M=5$). Each panel: ground truth (*left*), prediction (*center*), point-wise error (*right*).

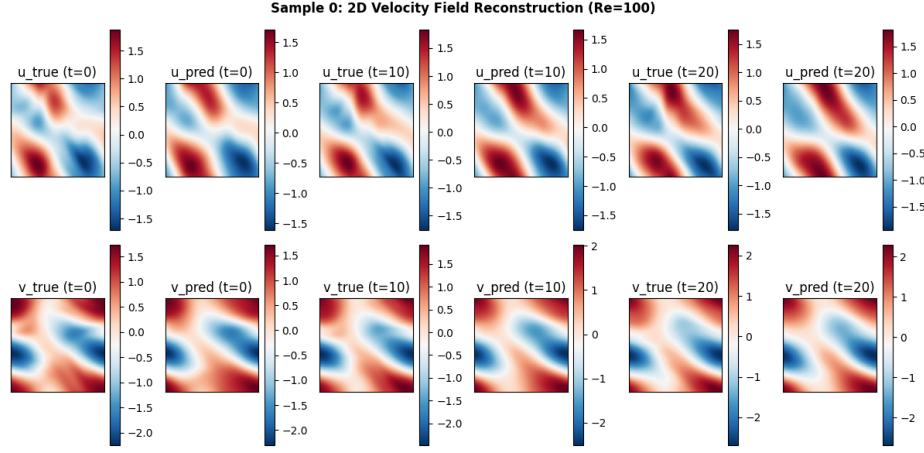


(a) TEMPO(POD)

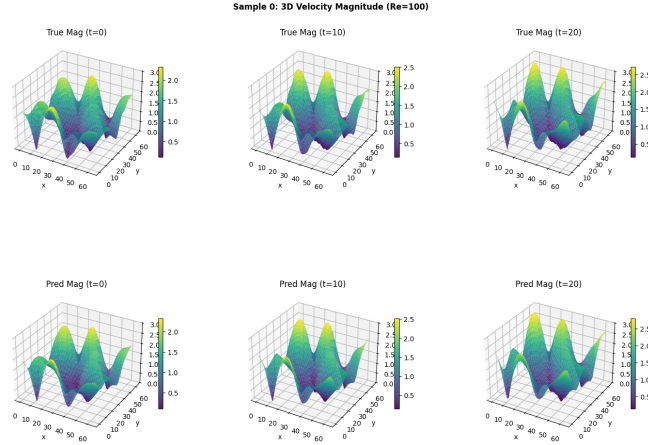
(b) TEMPO(NEURALPOD)

Figure 11: Space-time reconstructions on 2D Darcy flow ($M=5$).

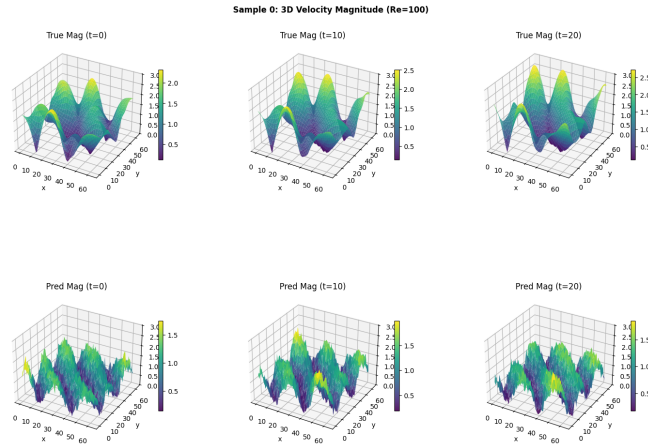
TEMPO reconstructions (2D Navier-Stokes). Figure 12 shows representative predictions from TEMPO(POD) and TEMPO(NeuralPOD) on 2D Navier-Stokes ($M=4$); EM convergence is shown in Figure 13.



(a) TEMPO(POD): ground truth (*left*), prediction (*center*), point-wise absolute error (*right*).



(b) TEMPO(POD) — 3D vorticity field.



(c) TEMPO(NeuralPOD) — 3D vorticity field.

Figure 12: 2D Navier-Stokes ($M=4$): TEMPO(POD) prediction (*top*); TEMPO(POD) and TEMPO(NeuralPOD) 3D vorticity (*middle, bottom*).

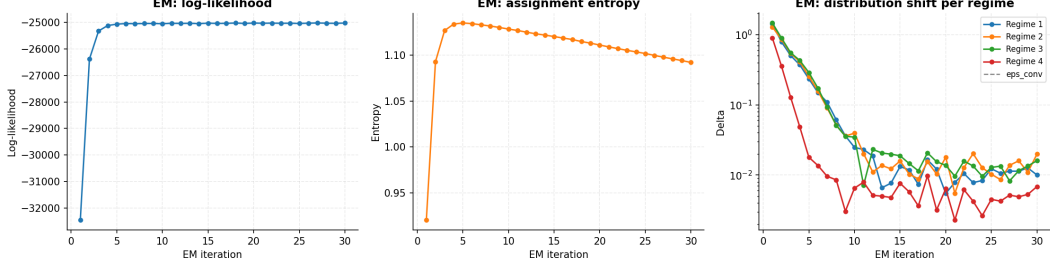


Figure 13: TEMPO(POD) EM convergence on 2D Navier-Stokes ($M=4$).

A.4 Algorithms

Algorithm 1 NEURALBASIS

Require: weighted snapshots $\{s^{(i)}, \gamma_{im}\}_{i=1}^N$, tolerance τ

Ensure: basis $\phi_0^{(m)}$, $\{\phi_k^{(m)}, \{\lambda_k^{(m,i)}\}_{i=1}^N\}_{k=1}^{K_m}$

- 1: $\bar{s}^{(m)} \leftarrow \frac{1}{N_m} \sum_{i=1}^N \gamma_{im} s^{(i)}$, $N_m = \sum_{i=1}^N \gamma_{im}$ ▷ responsibility-weighted mean trajectory
 - 2: Train $\phi_0^{(m)}$ by minimizing $\sum_j w_j (\phi_0^{(m)}(y_j) - \bar{s}_j^{(m)})^2$
 - 3: $r^{(0,i)} \leftarrow s^{(i)} - \phi_0^{(m)}(y)$ for all i ; $k \leftarrow 0$
 - 4: $V_m \leftarrow \sum_{i=1}^N \gamma_{im} \|s^{(i)} - \bar{s}^{(m)}\|_w^2$ ▷ initial unmodeled variance
 - 5: **while** $\sum_{i=1}^N \gamma_{im} \|r^{(k,i)}\|_w^2 \geq \tau V_m$ **and** $k < K_{\max}$ **do**
 - 6: Jointly minimize over $\phi_{k+1}^{(m)}$ and $\{\lambda_{k+1}^{(m,i)}\}_{i=1}^N$:

$$\sum_{i=1}^N \gamma_{im} \sum_j w_j \left(\lambda_{k+1}^{(m,i)} \phi_{k+1}^{(m)}(y_j) - r_j^{(k,i)} \right)^2$$
 - 7: $r^{(k+1,i)} \leftarrow r^{(k,i)} - \lambda_{k+1}^{(m,i)} \phi_{k+1}^{(m)}(y)$ for all i
 - 8: $k \leftarrow k + 1$
 - 9: **end while**
 - 10: **return** $K_m \leftarrow k$, $\phi_0^{(m)}$, $\{\phi_k^{(m)}, \{\lambda_k^{(m,i)}\}_{i=1}^N\}_{k=1}^{K_m}$
-

Algorithm 2 TEMPO

Require: $\mathcal{D} = \{u_0^{(i)}, \kappa^{(i)}, s^{(i)}\}_{i=1}^N$; $M, \varepsilon_s, \varepsilon_l, \varepsilon_p, \varepsilon_c, \tau$
Ensure: Regime bases $\{\Phi_m^*\}$, amplitudes $\{\Lambda_m^*\}$, responsibilities $\{\gamma_{im}^*\}$, weights $\{\pi_m^*\}$

Initialization

- 1: Compute global POD on $\{s^{(i)}\}$; project to $\alpha^{(i)} \in \mathbb{R}^{d_0}$ for all i
- 2: Run GMM-EM on $\{\alpha^{(i)}\} \rightarrow \gamma_{im}^{(0)}, \pi_m^{(0)}, \mu_m^{(0)}, \Sigma_m^{(0)}$
- 3: **for** $m = 1, \dots, M$ **do**
- 4: $\Phi_m^{(0)}, \Lambda_m^{(0)} \leftarrow \text{NEURALBASIS}(\{s^{(i)}, \gamma_{im}^{(0)}\}, \tau)$ $\triangleright$ or weighted SVD for TEMPO(POD), see §3.1
- 5: **end for**
- 6: $\sigma^2 \leftarrow (2N)^{-1} \sum_{i,m} \gamma_{im}^{(0)} \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2$ (14)

EM iterations

- 7: **repeat**
- 8: *E-step*
- 9: **for** all $i = 1, \dots, N$ and $m = 1, \dots, M$ **do**
- 10: Compute $\hat{s}_m^{(i)}$ via (4) using $\Phi_m^{(q-1)}$ and $\Lambda_m^{(q-1)}$
- 11: $\gamma_{im}^{(q)} \leftarrow \text{Eq. (15)}$
- 11: **end for**
- 12: *M-step: mixture weights*
- 13: $\pi_m^{(q)} \leftarrow N_m^{(q)} / N$ for all m (16)
- 14: *M-step: bases*
- 15: **for** $m = 1, \dots, M$ **do**
- 16: Compute $\mu_m^{(q)}$ (18), $\Sigma_m^{(q)}$ (19), $\Delta_m^{(q)}$ (20)
- 17: **if** $\Delta_m^{(q)} < \varepsilon_s$ **then**
- 18: $\Phi_m^{(q)} \leftarrow \Phi_m^{(q-1)}$ SKIP
- 19: **else if** $\Delta_m^{(q)} < \varepsilon_l$ **then** INCREMENTAL
- 20: Compute $r_m^{(K_m)}$ under updated $\gamma^{(q)}$; $V_m^{(q)} \leftarrow \sum_i \gamma_{im}^{(q)} \|s^{(i)} - \bar{s}_m^{(q)}\|_w^2$
- 21: **if** $\sum_i \gamma_{im}^{(q)} \|r_m^{(K_m, i)}\|_w^2 \geq \tau \cdot V_m^{(q)}$ **then**
- 22: Train mode $K_m + 1$; $K_m \leftarrow K_m + 1$
- 23: **end if**
- 24: **if** $\|\phi_{K_m}^{(m)}\|_w^2 \cdot \sum_{i=1}^N \gamma_{im}^{(q)} (\lambda_{K_m}^{(m, i)})^2 < \varepsilon_p$ **then**
- 25: $K_m \leftarrow K_m - 1$
- 26: **end if**
- 27: **else** FULL RERUN
- 28: $\Phi_m^{(q)}, \Lambda_m^{(q)} \leftarrow \text{NEURALBASIS}(\{s^{(i)}, \gamma_{im}^{(q)}\}, \tau)$
- 29: **end if**
- 30: **end for**
- 31: **until** $\max_m \Delta_m^{(q)} < \varepsilon_c$
- 32: **return** $\Phi_m^* \leftarrow \Phi_m^{(q)}, \Lambda_m^* \leftarrow \Lambda_m^{(q)}, \gamma_{im}^* \leftarrow \gamma_{im}^{(q)}, \pi_m^* \leftarrow \pi_m^{(q)}$
